| Problem Chosen | 2023 MCM/ICM Summary Sheet | Team Control Number |
|---|---|---|
| B | | 2305501 |

# Human and Nature: More Productive Ecological Policy in Maasai Mara


The rapid development of the economy has brought increased attention to the interaction between economic development and the ecological environment. In Kenya, game reserves were established with the primary goal of protecting wildlife and natural resources. However, amendments to legislation have been added over time to address gaps in governance, funding, and penalties for offenders.

To ensure coordinated economic development and ecological protection, the aim of this study is to promote production and lifestyle practices that minimize pollution and ecological damage during moderate economic development. Through a combination of vertical comparison and mathematical modeling, we explore the interaction between economic development and the ecological environment in Kenya.

This paper analyzes the main factors that influence the interaction between economic development and the ecological environment in Kenya and proposes a framework for the reasonable development of various indices. By identifying and addressing the key drivers of environmental degradation and promoting sustainable economic growth, we can ensure a more harmonious and sustainable relationship between economic development and the ecological environment.

- For question 1: We first use Maximum Likelihood Method and Iso Clustering Model to achieve supervised and unsupervised classification of map data, and then divide three areas: Cropping Area, Livestock Area, and Tourism Area. Cropping area, livestock area, and tourism area. Due to each region can adopt different policies. Through data collection, we listed and quantified the specific implementation plans of six kinds of policies.
- For question 2: In this question, we first adopted the multiple linear regression model to predict the short-term impact of the implementation of various programs and select appropriate strategies through various indicators in the Maasai Mara region from 2017 to 2021. Then, we used TOPSIS comprehensive evaluation model to further explore these data sets, and obtained the rationality of adopting the above policies in the short term.
- For question 3: In order to realize long-term prediction and promote and apply policies to other wildlife reserves, we adopted the Random Forest Model and the LSTM Model respectively to train the acquired data sets. By observing the accuracy of the prediction results of the test set, the prediction curve and the loss curve can realize the promotion of the best policy.



**Keywords:** Human and Wildlife, Reverse Management, Classification, Evaluation Model, Time Series Forecasting

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# 1 Introduction

## 1.1 Problem Restatement

1. Background

Kenya's wildlife preserves were originally created primarily to protect wildlife and other natural resources. Kenya's parliament passed the Wildlife Conservation and Management Act, 2013 to provide more equitable sharing of resources, as well as to allow alternative, community-based management efforts. Kenya has since added amendments to address gaps in the legislation to provide more clear governance, finance, and penalties for violators.

2. Requirement

Focusing on one large game preserve, the Maasai Mara, our task is to determine alternate ways to manage the resources within and outside the current boundaries of the park. Specifically, we should:

- Consider and recommend specific policies and management strategies for different areas within the current preserve that will protect wildlife and other natural resources while also balancing the interests of the people who live in the area. These policies and strategies should help mitigate the impacts of lost opportunities experienced by the people who live near the preserve, as well as minimize negative interactions between animals and the people attracted to the preserve.
- Develop and describe a methodology to determine which policies and management strategies will result in the best outcomes. Our report should discuss how to rank and compare outcomes from your methodology. T he paper should include descriptions and analyses of the models used to predict the interactions between animals and people, as well as the resulting economic impacts in the area within and around the preserve.
- Given our proposed plan, provide predictions about the long-term trends that will result from our recommendations. Analyze and provide estimates of the certainties and impacts of the possible long-term outcomes. We need to describe how our approach could be applied to other wildlife management areas. Finally, provide a two-page non-technical report for the Kenyan Tourism and Wildlife Committee discussing our proposed plan and its value for the preserve.

## 1.2 Our Work

**Problem 1**

We will recommend specific policies and strategies for different areas within current preserve, considering people and wildlife at the same time. We utilized classification

methods to perform supervised and unsupervised classification of map data. This allowed us to divide the area into different distinct zones: Cropping Area, Livestock Area, and Tourism Area. Each of these areas has unique characteristics that require tailored policies to achieve optimal outcomes. To develop specific policy plans, we collected data and quantified some policy options that can be implemented in each respective zone. By identifying the appropriate policies for each area, we can effectively manage the ecological environment, protect wildlife, and promote sustainable economic growth.

**Problem 2**

We will develop a methodology to rank and compare our ploicies and management strategies, hoping to pick the most suitable policies and management strategies which will result in best outcomes. To address this question, we utilized several models to predict the short-term impacts of different programs implemented in the Maasai Mara region from 2017 to 2021. The model enabled us to identify appropriate strategies based on various indicators.

**Problem 3**

According to the recommendations we have chosen in Problem 2, we will provide predictions about its long-term trends along with estimates of certainties and impacts of the long-term outcomes. Meanwhile, we will describe how our approach could be applied to other wildlife management areas. To ensure the long-term effectiveness of the policies and to promote their applicability to other wildlife reserves, we utilized some advanced models to train the acquired datasets.

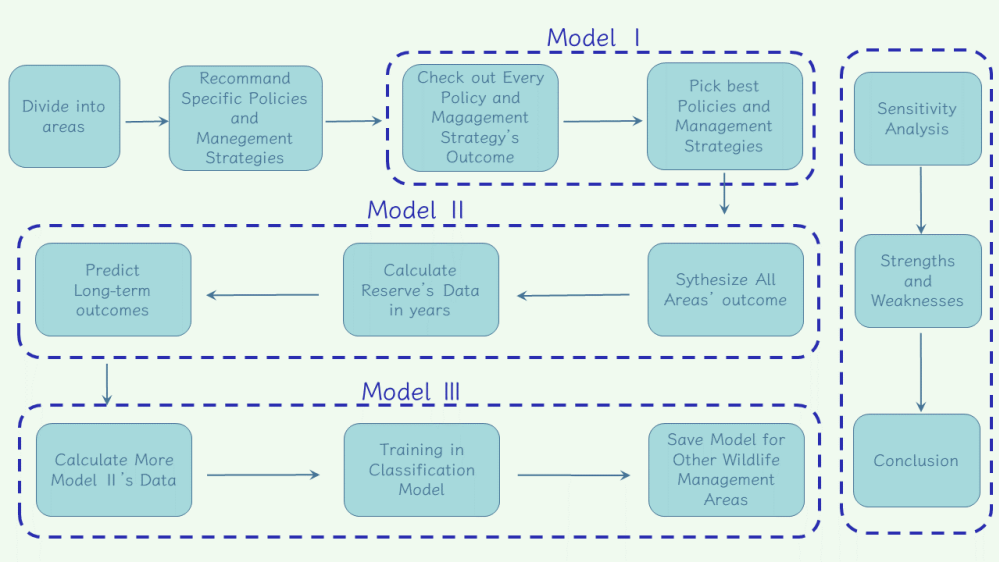



Figure 1: Our Work

# 2 Preparation of the Models

## 2.1 Assumptions

- This paper assumes that the interaction between policies with sufficiently low correlation is negligible. This facilitates policy selection and data processing and reduces the cost of data processing.
- This paper assumes that the ecological and economic conditions of Maasai Mara conform to the universal rules in the UN report. This can be convenient to quantify and measure the specific implementation strategy of a certain policy, so as to give a forecast model.
- This paper assumes that the changes of various ecological and economic indicators in adjacent years will not be too rapid to prevent the phenomenon of sudden increase and decrease of data.

## 2.2 Notation

Table 1: Notations

| Symbol | Definition |
|---|---|
| $a_f$ | The proportion of farming area |
| $c_i$ | Crop yield increase rate |
| $a_p$ | Pasture area |
| $s_{\text{num}}$ | Number of species |
| $\rho_s$ | Species density |
| $\rho_l$ | Livestock density |
| $i_{\text{num}}$ | Animal injury incidents |
| $t_{\text{in}}$ | Tourism income |
| $t_{\text{num}}$ | Number of tourists |
| $G$ | The increasing rate of GDP |
| $F$ | The coverage rate of forest |
| $\alpha$ | The effect index of policies in LSTM |
| $\beta$ | Relative closeness to the ideal solution |

# 3 Model Construction

## 3.1 Management of Regions and Policies

### 3.1.1 Region-Segmentation

1. Model Introduction

The multivariate classification method of ArcGIS was adopted to divide the three regions of Maasai Mara.

The purpose of area partitioning is to assign each pixel in the study area to a class or directory. Examples of classes or catalogs include farmland area, livestock density in pasture, and nature reserve area.

There are two types of classification: supervised and unsupervised.

In supervised classification, **Maximum Likelihood** is used to identify it by surrounding it with a face (or faces) on the map, sampling the elements, and continuing the process until there are enough elements to represent the class and all the classes in the data have been identified. Each set of elements is considered a class, and the surfaces surrounding that class are the training samples. After the training samples were identified, multivariate statistical data were calculated to establish relationships within and between classes. Statistics are stored in feature files.

In the unsupervised classification, **Iso Clustering** is used to calculate the evenly distributed class mean in the data space, and then the minimum distance technology is used to iterate the remaining pixels for aggregation. The mean value is recalculated in each iteration, and pixels are reclassified according to the new mean value obtained, so that each position is aggregated into one of the specified number of groups or clusters. Multivariate statistical calculations of the input bands are used to determine which class each position is assigned to. The value of the pixel band in the cluster determines that each cluster is statistically independent of the others. At the same time, the statistics used to establish the clustering definition will also be stored in the characteristics file.

A class corresponds to a meaningful location grouping. Each location can be characterized by a value set or a value vector, a value corresponding to each variable, or a band input in the analysis. Each position can be displayed as a point in the multidimensional attribute space of the axis corresponding to the variable represented by each input band. A class or cluster is a set of points in such a multidimensional attribute space. If the properties (vectors of band values) of two locations are similar, the two locations will belong to the same class or cluster. Multiband grids and each single band grid can be used as inputs for multivariate statistical analysis.

2. Model Building

- Analyze, process and classify the collected data

  Kenya GIS Data collected from the World Resources Institute website, Select Areas of cropland in Kenya, Livestock density in Kenya's rangelands, Protected areas in Kenya and Predicted mammal diversity, the total number of mammal species in Kenya, Tourist accommodations in Kenya, Average inequality of per capita expenditures in Kenya (also known as the GINI Index, Major towns in Kenya, etc.) are drawn in arcmap and the following figure is obtained.

  By referring to *IIED Wildlife & Development Series.Wildlife and People: Conflict and Conservation in Maasai Mara*, the division of Maasai Mara in Kenya roughly corresponds to Kenya's pillar industries: planting industry, animal husbandry and tourism service industry (Direction 1, 2, 3) in the figure below.

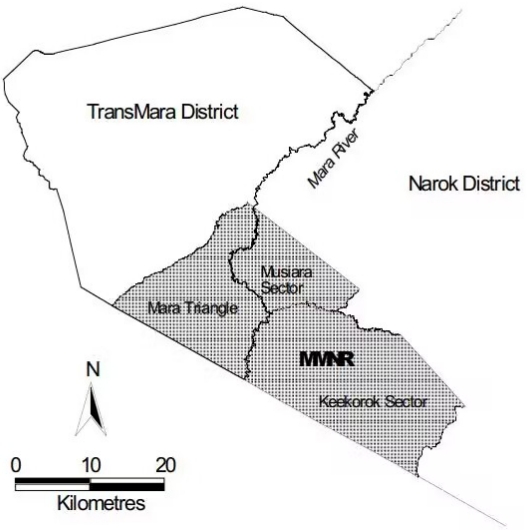



Figure 2: Region Segmentation

By looking at data from the Masai Mara Game Reserve, plantation areas were associated with farmland size, crop yields, overlapping elephant and rhino habitats, and animal injuries. Livestock farming is related to the density of livestock in the pastures, milk production, wildlife density in the pastures of Kenya; Tourism is related to species diversity, the size of nature reserves, hotels and the distribution of major cities.

- Multivariate classification of data

The elements of hotels, major cities and airports in Masai Mara were obtained by **Iso Unsupervised Clustering** to obtain special files. Then, the indicator legend elements related to planting industry and animal husbandry were sampled on the map and represented as classes. As training samples, the feature files were generated and combined with the previous one as the feature files of the three classes. With the **Maximum Likelihood Method**, enter the raster band and established feature file with the default value of 0.0 reject_fraction, indicating that each pixel will be classified, and specify that all classes have the same prior probability using EQUAL mode. Finally, the output is carried out to obtain the fine division of the three areas, which is supported by the data.

- Sorting and post-processing

We use reclassification to merge subclasses and use block statistics to remove noise from the result.

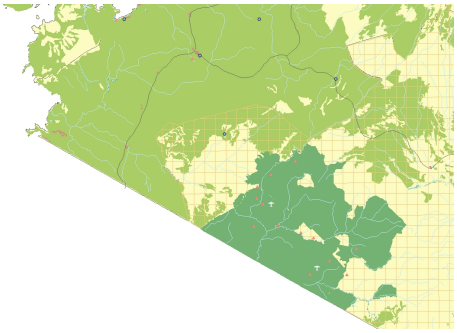


Figure 3: Maasai Mara

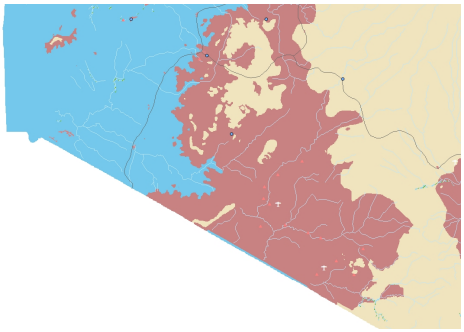


Figure 4: Area've been Devided

### 3.1.2 Executable Relational Policies

**Direction 1**

Cropping areas are associated with farmland size, crop yields, overlapping elephant and rhino habitats, and animal injuries.

- **Policy 1**: Limit population growth and land reclamation activities

  A landmark new report from the Intergovernmental Science-Policy Platform on Biodiversity and Ecosystem Services (IPBES) indicates controlling population growth and reclamation activities to limit farmland area to **90**% of its current size[1].

- **Policy 2**: Adopt eco-friendly planting methods

  Adopting organic ecology improves crop yields up to **8.4**% per year, according to *Coherent Market Insights*[2].

**Direction 2**

Livestock farming is related to the density of livestock in the pastures, milk production, wildlife density in the pastures of Kenya.

- **Policy 3**: Wildlife habitats and seasonal migration routes should be taken into account

  By investigating the migratory routes of wild animals in the map, the area of pasture can be reduced to **92**% by taking into account the migratory routes of wild animals.

- **Policy 4**: Reduce contact between wildlife and humans to strive for **NO** casualties

  An example of this can be seen in the Kavango Zambezi Transfrontier Conservation Area in Southern Africa, where an integrated approach to managing human-wildlife conflict has led to a **95**% reduction in livestock killings, resulting in zero retaliatory killings[3] of lions in 2016.

| | Elephant | | Herbivores | | Predators | | Total |
|---|---|---|---|---|---|---|---|
| Period | Deaths | Injuries | Deaths | Injuries | Deaths | Injuries | |
| 1986-1990 | 1 | 2 | 1 | 2 | 0 | 2 | 8 |
| 1991-1995 | 9 | 5 | 0 | 9 | 0 | 5 | 28 |
| 1996-2000 | 11 | 7 | 1 | 5 | 2 | 20 | 46 |
| Total | 21 | 14 | 2 | 16 | 2 | 27 | 82 |

**Table 5. Numbers of people attacked by wildlife in TransMara District between 1986 and 2000.**

Figure 5: Selectable Policies

**Direction 3**

Tourism is related to species diversity, the size of nature reserves, hotels and the distribution of major cities.

- **Policy 5**: Infrastructure and road construction

  Due to the damage to the land caused by the tire tracks of the tour group's cross-country convoy in the reserve and disturbance to the animals, fragmentation is a landscape level process in which a specific habitat is progressively subdivided into. Preventing habitat fragmentation could boost animal populations by **25**%[4].

- **Policy 6**: Limit tourist numbers

  Limiting the number of visitors helps to protect the living environment of wild animals and prevent excessive disturbance. This implies that a **1**% increase in population growth, biodiversity loss will increase by **42**%[5].

## 3.2 Strategies of Methodology

### 3.2.1 Multivariable Linear Regression Model

1. Model Introduction

**Multivariable Linear Regression Model** is a statistical technique used to analyze the relationship between multiple independent variables and a single dependent variable[6]. It is a powerful tool used to predict the values of the dependent variable based on the values of the independent variables. This model is commonly used in various fields such as

finance, economics, and social sciences to make predictions, identify trends, and analyze the impact of different variables.

The Multivariable Linear Regression Model is a mathematical equation that describes the relationship between the dependent variable and the independent variables. The model takes the following form:

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \ldots + \beta_n X_n + \varepsilon \tag{1}$$

where $Y$ is the dependent variable, $\beta_0$ is the intercept, $\beta_1, \beta_2, \ldots, \beta_n$ are the regression coefficients that represent the effect of each independent variable on the dependent variable, $X_1, X_2, \cdots, X_n$ are the independent variables, and $\varepsilon$ is the error term.

The regression coefficients show how much the dependent variable changes for a one-unit change in the independent variable while holding all other independent variables constant. The Multivariable Linear Regression Model uses a least-squares method to estimate the regression coefficients.

The objective of this method is to find the values of $\beta_0, \beta_1, \beta_2, \cdots, \beta_n$ that minimize the sum of the squared differences between the observed values of the dependent variable and the predicted values based on the independent variables. This is done by calculating the partial derivatives of the sum of squares with respect to each regression coefficient and setting them equal to zero.

## 2. Model Building

The aim of this study is to develop a multiple linear regression model to predict two dependent variables based on nine independent variables. The data set provided includes ten variables, where the first nine are independent variables and the last two are dependent variables. The independent variables are related to ecological and economic factors, while the dependent variables are the forest coverage rate and GDP.

The model developed will help to understand the relationship between the independent and dependent variables and to predict the values of the dependent variables based on the values of the independent variables.

By quantifying the **proportion of farming area** ($a_f$), **crop yield increase rate** ($c_i$), **pasture area** ($a_p$), **number of species** ($s_{\text{num}}$), **species density** ($\rho_s$), **livestock density** ($\rho_l$), **animal injury incidents** ($i_{\text{num}}$), **tourism income** ($t_{\text{in}}$), and **number of tourists** ($t_{\text{num}}$), we can measure the six policies mentioned above. Using multiple linear regression models for the above policies, we get

$$\begin{aligned} G \;=\; & 0.4502a_f + 0.0291c_i + 0.0858a_p + 0.0343s_{\text{num}} - 0.0253\rho_s + 0.0124\rho_l \\ & + 0.0132i_{\text{num}} - 0.0072t_{\text{in}} + 0.0004t_{\text{num}} + 6.1597 \end{aligned} \tag{2}$$

where $G$ represents the **increasing rate of GDP**.

$$\begin{aligned} F &= 0.3883a_f + 1.3104c_i + 3.5413a_p + 0.0054s_{\text{num}} + 0.0142\rho_s + 0.0359\rho_l \\ &\quad - 0.0138i_{\text{num}} - 0.0936t_{\text{in}} - 0.0045t_{\text{num}} - 1.0306 \end{aligned} \tag{3}$$

where $F$ represents the **coverage rate of forest**.

The predicted results are as follows:

Table 2: Prediction of Linear Regression Model

| Policies | GDP Increase Rate | Forest Coverage Rate |
|---|---|---|
| 1 | 5.75% | 6.32% |
| 2 | 6.08% | 6.35% |
| 3 | 5.34% | 6.29% |
| 4 | 5.79% | 6.40% |
| 5 | 6.92% | 6.33% |
| 6 | 5.74% | 6.28% |

We know from the table that in order to give consideration to ecological environment and economic benefits, we should give priority to **Policy 2**, **Policy 4** and **Policy 5**.

### 3.2.2 TOPSIS Model

1. Model Introduction

In addition to the multiple linear regression model, we also used **TOPSIS Comprehensive Evaluation Model** to evaluate the above policies.

The Technique for Order Preference by Similarity to Ideal Solution (TOPSIS) is a multi-criteria decision-making method used to evaluate the best alternative among a set of given alternatives based on a set of criteria. It was developed by Hwang and Yoon in 1981[7] and has since been widely used in various fields, including engineering, management, economics, and environmental studies.

The TOPSIS model works by comparing the alternatives to a set of ideal and negative ideal solutions based on the criteria. The ideal solution represents the best possible values for all criteria, while the negative ideal solution represents the worst possible values for all criteria. The distance between each alternative and the ideal and negative ideal solutions is then calculated using a distance metric, such as Euclidean distance or Manhattan distance.

The mathematical representation of the TOPSIS model is as follows:

Let there be $m$ alternatives and $n$ criteria. The performance of each alternative on each criterion is represented by a decision matrix $X$ with dimensions $m, x, n$, where $x_{ij}$ denotes the performance of alternative $i$ on criterion $j$.

The first step is to normalize the decision matrix to eliminate the scale differences among the criteria. The normalized decision matrix $R$ with dimensions $m, x, n$ is calculated as:

$$r_{ij} = \frac{x_{ij}}{\sqrt{\sum_{j=1}^{n}(x_{kj}^2)}} \tag{4}$$

where $k = 1, 2, ..., m$.

Next, the weighted normalized decision matrix V with dimensions $m, x, n$ is calculated as:

$$v_{ij} = w_{ij} \cdot r_{ij} \tag{5}$$

for $j = 1, 2, ..., n$.

where $w_{ij}$ is the weight of criterion $j$.

Then, the ideal solution $A^+$ and negative ideal solution $A^-$ with dimensions $1, x, n$, are calculated as:

$$\begin{aligned} A_j^+ &= \max(v_{ij}) \\ A_j^- &= \min(v_{ij}) \end{aligned} \tag{6}$$

for $i = 1, 2, ..., m$ and $j = 1, 2, ..., n$.

The distance between each alternative $i$ and the ideal and negative ideal solutions is calculated using a distance metric $d$ as follows:

$$\begin{aligned} d_i^+ &= d(v_i, A^+) = \sqrt{\sum_{i=1}^{m}(v_{ij} - A_j^+)^2} \\ d_i^- &= d(v_i, A^-) = \sqrt{\sum_{i=1}^{m}(v_{ij} - A_j^-)^2} \end{aligned} \tag{7}$$

Finally, the relative closeness to the ideal solution $C^+$ for each alternative $i$ is calculated as:

$$C_i^+ = \frac{d_i^-}{(d_i^+ + d_i^-)} \tag{8}$$

for $i = 1, 2, ..., m$.

The alternative with the highest value of $C_i^+$ is considered to be the best alternative.

2. Model Building

Firstly, we determine the decision matrix and record the attribute values of all candidate policies in the decision matrix. The decision matrix contains the above six policies and various ecological and economic indicators, in which each element represents the score of the corresponding policy on a certain attribute.

Then the obtained data will be normalized to prevent the data difference from being too large after the establishment of the model. For the maximum attribute, the higher the score, the higher the score. For the minimum attribute, the lower the score, the higher the score for the policy.

The positive ideal solution (that is, the policy with the highest score on each attribute) and the negative ideal solution (that is, the policy with the lowest score on each attribute) are then determined, thus calculating the Euclidean distance between each policy and the positive and negative ideal solutions.

For each scheme, calculate the ratio $\beta$ of the distance to the positive ideal solution to the negative ideal solution. The closer the ratio is to 1, the better the scheme is, and the worse it is.

Finally, the scheme is ranked as below:

Table 3: Prediction of TOPSIS Model

| Policies | $\beta$ |
|---|---|
| 1 | 0.394006386 |
| 2 | 0.427943910 |
| 3 | 0.398977047 |
| 4 | 0.410027614 |
| 5 | 0.434297480 |
| 6 | 0.403487545 |

By choosing options closer to 1, the model tells us that **Policy 2**,**Policy 4**, and **Policy 5** should be adopted preferentially. This is quite similar to the result obtained by using the linear regression model.

## 3.3 Prediction of the Future

### 3.3.1 Random Forest Model

1. Model Introduction

**Random Forest Model** is an ensemble learning technique that is widely used for predictive analysis. It is based on the concept of decision trees, which are used to make a series of decisions based on input data[8]. In Random Forest Model, multiple decision trees are created, and the results are combined to produce a final prediction. This technique is particularly useful when dealing with complex data, as it is able to handle high-dimensional datasets, missing values, and noisy data.

Random Forest Model is based on the concept of decision trees. In this model, multiple decision trees are created, and their results are combined to produce a final prediction. The following mathematical notation is used to describe the Random Forest Model:

Suppose we have a training dataset $D = \{(x_1, y_1), (x_2, y_2), \cdots, (x_n, y_n)\}$, where

$$x_i = (x_{i1}, x_{i2}, \cdots, x_{im}) \tag{9}$$

$x_i$ is a vector of m independent variables, and $y_i$ is the corresponding dependent variable. Let $T$ be the number of decision trees, and let $t_i$ be the $i$th decision tree.

Each decision tree $t_i$ is constructed using a random subset of the available features. Let the number of features to be selected be $k$. We randomly select $k$ features from the $m$ features in the dataset. The decision tree $t_i$ is then constructed using these $k$ features and the training data.

In the bootstrapping step, multiple samples are drawn with replacement from the training dataset. Let $D_i$ be the $i$th sample, where $D_i$ is a subset of the training dataset with $n$ observations. The decision tree $t_i$ is constructed using $D_i$.

Finally, in the voting step, the predictions from all the decision trees are combined. Let the prediction from the $i$th decision tree be $y_i(t_i, x_i)$, and let the final prediction be

$$yf(x) = F(y_1(t_1, x), y_2(t_2, x), \ldots, y_T(t_T, x)) \tag{10}$$

where $F$ is a function that combines the predictions.

2. Model Building

In the second question, TOPSIS model is selected for processing, and the data from 2017 to 2020 are also processed by policies. After applying TOPSIS model, comprehensive indicators are obtained and corresponding policies are selected. In order to extend the policies to more areas, random forest model is applied for classification, and all data

indicators are taken as parameters. The choice of decision builds the model for the outcome. After training, we open source the model to help the government make decisions.

We randomly selected these data samples and features, and randomly selected some samples from the training data, and then randomly selected some features from the features of each sample, namely self-sampling and random subspace sampling. The goal is to reduce correlations between samples and reduce the number of features.

The key step is to train multiple decision trees. Through the above steps, we can obtain multiple training subsets. For each subdata set, a decision tree model is trained and the prediction error is gradually reduced by iterative segmentation. For each sample in the test set, the prediction is made through all the trained decision trees, and then the average or majority vote is taken to get the final prediction result. In order to improve its prediction ability and generalization ability, we have carried out noise reduction and de-noising processing on the data.

By comparing the 2017-2021 data forecast with the actual situation, we get

Table 4: LSTM Prediction Model

| Year | Prediction | Reality |
|---|---|---|
| 2017 | 2 3 5 | 2 3 5 |
| 2018 | 1 4 6 | 1 4 6 |
| 2019 | 1 4 5 | 1 4 5 |
| 2020 | 2 3 6 | 2 3 6 |
| 2021 | 2 4 5 | 2 4 5 |

It is found that the prediction ability of this model is very accurate and can be used in other areas of wildlife reserve management.

### 3.3.2 Long Short-Term Memory Model

1. Model Introduction

**LSTM** (Long Short-Term Memory) is a type of recurrent neural network (RNN) architecture [9] that is widely used in natural language processing, speech recognition, and many other sequence prediction tasks. The key feature of LSTM is its ability to handle long-term dependencies in a sequence by selectively remembering or forgetting information from previous time steps. In an LSTM model, information flows through a sequence of "memory cells" that are interconnected through gates. There are three types of gates in an LSTM model:

Forget gate: controls the amount of information from the previous memory cell to be discarded.

Input gate: controls the amount of new information to be added to the current memory cell.

Output gate: controls the amount of information from the current memory cell to be used as the output.

The calculation of each gate is done by using sigmoid and tanh activation functions. The sigmoid function outputs a value between 0 and 1, which represents the degree of forgetting or retaining information. The tanh function outputs a value between -1 and 1, which represents the amount of new or stored information to be added to the memory cell. The LSTM model takes in a sequence of input vectors, and for each time step, it computes the output vector based on the current input and the previous hidden state vector. The output vector can be used for various tasks such as classification, regression, or generation of new sequences.

2. Model Building

Since the three regions have selected a policy respectively, we add the comprehensive results obtained by TOPSIS in the three regions as the effect of our policy. After that, we combined the data from 2017 to 2020 with TOPSIS results to obtain the aggregate data that would have been obtained in previous years if policies had been implemented that were appropriate for the situation. Then the LSTM time series prediction model is used to predict the comprehensive data from 2022 to 2030.

First of all, we preprocessed the ecological and economic data required by LSTM network, including text word segmentation, numerical standardization or normalization.

Next we built the LSTM network structure and used the labeled data set to train the LSTM model in order to minimize the difference between the model prediction and the actual label. When training the model, LSTM uses the back propagation algorithm to adjust the parameters of the neural network to minimize the prediction error.

Finally, the new data is input into the LSTM model, and the prediction results are generated through the network structure. Since the three regions have selected a policy respectively, we add the comprehensive results obtained by TOPSIS in the three regions as the effect of our policy $\alpha$.

Table 5: LSTM Prediction Model

| Year | 2022 | 2023 | 2024 | 2025 | 2026 | 2027 | 2028 | 2029 | 2030 |
|---|---|---|---|---|---|---|---|---|---|
| $\alpha$ | 0.589 | 0.738 | 0.667 | 0.648 | 0.725 | 0.589 | 0.608 | 0.682 | 0.601 |

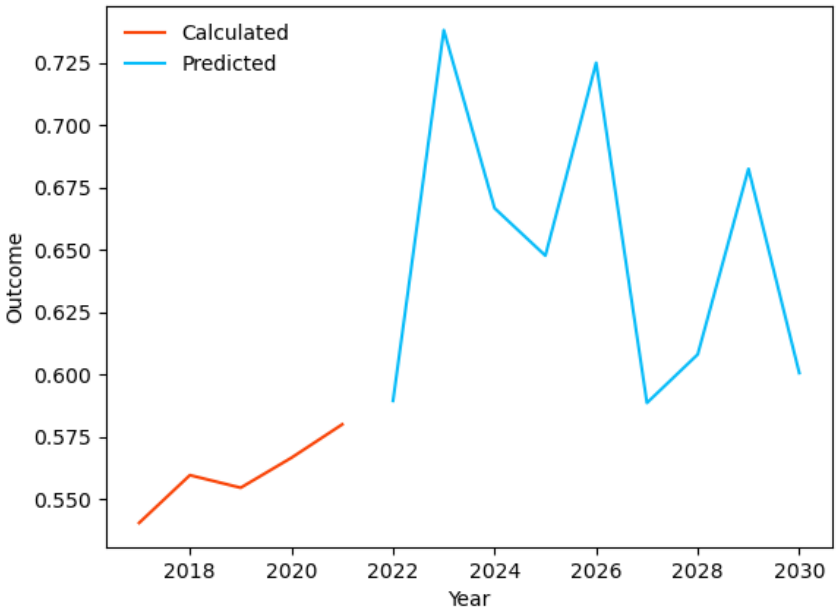



Figure 6: Prediction Curve

with the lost curve is presented below:

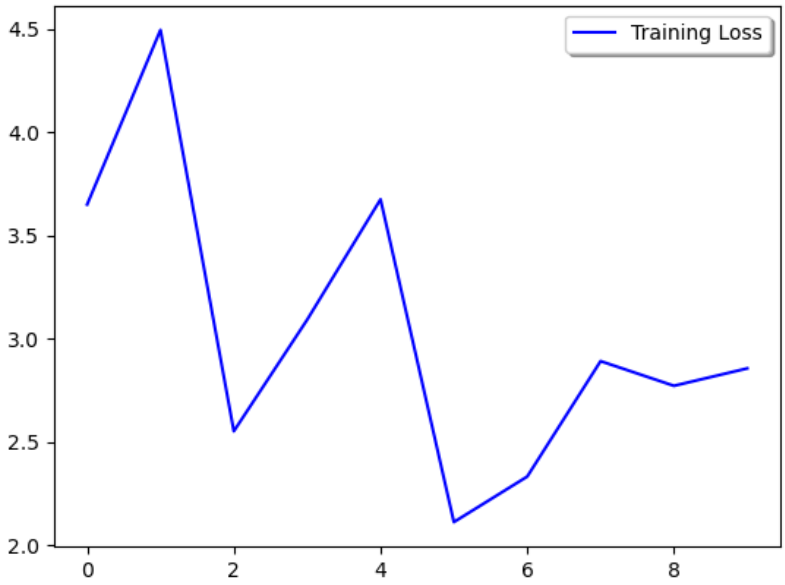



Figure 7: Loss Curve

# 4 Model Evaluation

## 4.1 Sensitivity Analysis

By analyzing the proportion of farming area, crop yield increase rate, pasture area, number of species, species density, livestock density, animal injury incidents, tourism income, and number of tourists data and dependent variable GDP growth rate and forest cover ratio using **Grey Correlation Model**, we can conclude:

1. Farming area, crop yield increase rate, pasture area, livestock density and other planting industry related changes have little impact on local ecology and economic increase rate.
2. The number of species, species density is closely related to the change of local ecological environment.
3. Tourism income, number of tourists and other tourism indicators basically directly reflect the change of economic conditions.

Table 6: Grey Correlation

| Correlation | $a_f$ | $c_i$ | $a_p$ | $s_{\text{num}}$ | $\rho_s$ | $\rho_l$ | $i_{\text{num}}$ | $t_{\text{in}}$ | $t_{\text{num}}$ |
|---|---|---|---|---|---|---|---|---|---|
| $G$ | 0.323 | 0.297 | 0.393 | 0.401 | 0.423 | 0.330 | 0.423 | 0.818 | 0.792 |
| $F$ | 0.431 | 0.317 | 0.458 | 0.793 | 0.762 | 0.294 | 0.401 | 0.501 | 0.560 |

## 4.2 Strengths and Weaknesses

Strengths:

- The model has strong scalability, whether adding a large number of data, adding appropriate indicators, or modifying specific contents of policies.
- In addition, TOPSIS Model has strong reusability and unified formula, without much modification. For the Random Forest Model, the data can be retained, which greatly reduces the running time. At the same time, reduce user operations, easy to use, a wider range of audiences.
- For the LSTM Model, the relevant data in the long interval can be retained during the prediction, which has a good effect on the long and short term prediction.

Weaknesses:

- The amount of data acquired is not enough, five years of data is still not enough to train a lot.
- LSTM Model may be volatile during Short-term forecasts.
- The weight of topsis model is defined by entropy weight, which is different from the real standard.

# 5 Conclusion

To summarize, after collecting and analyzing local data, we have identified the three policies with the most promising outcomes for wildlife management in the Masai Mara Nature Reserve. These policies are: adopting eco-friendly planting methods in cropping areas, reducing contact between wildlife and humans to minimize casualties in livestock farming areas, and constructing infrastructure and roads in tourism areas. These policies have been carefully selected to ensure that they align with local development goals while also protecting wildlife and maintaining ecological balance.

Our innovative approach to policy selection, which incorporates the Random Forest Model, has enabled us to extend the applicability of our policies to other wildlife management areas in Kenya and beyond. The Random Forest Model leverages machine learning algorithms to identify the most economically and ecologically viable policies for a given region, taking into account local conditions and characteristics. This approach greatly simplifies the selection process and can help conservationists identify the most effective policies for protecting wildlife and promoting sustainable development.

In addition, we have used the Long Short-Term Memory (LSTM) Model to predict the future development trends of policy implementation. This analysis reveals dynamic changes in various indicators and provides guidance for future policy adjustments. By continuously monitoring and adjusting our policies, we can ensure that our conservation efforts remain effective and sustainable over the long term.

FOSTER A COMMUNITY OF LIFE FOR MAN AND NATURE

# Policy Selection and Promotion of Maasai Mara Nature Reserve

FOR KENYA TOURISM &WILDLIFE COMMISSION

Just as the Chinese Taoism said, " For I am abstracted from the world，the world from nature，nature from the way，and the way from what is beneath abstraction", Chinese people have been pursuing harmonious coexistence between man and nature since ancient times.

In formulating policies for the Masai Mara Nature Reserve in Kenya, we insist that the goal of wildlife management is to protect wildlife, maintain ecological balance, and maximize economic benefits in the process. Firstly, we collected and analyzed all kinds of local data, and preliminarily wrote Limit population growth and land reclamation activities, Adopt eco-friendly planting methods, Wildlife habitats and seasonal migration routes should be taken into account, Reduce contact between wildlife and humans to strive for NO casualties, Infrastructure and road construction, Limit tourist numbers six policy.

We conducted multiple regression analysis, data testing and deep learning algorithm prediction on policies and indicators. After a series of rigorous demonstration, Adopt eco-friendly planting methods, Reduce contact between wildlife and humans to strive for NO casualties, Infrastructure and road construction are three management strategies that can effectively achieve this goal, which can not only protect the living environment of wildlife, ensure the per capita income and food output of Masai Mara, but also attract tourists, thus increasing local economic benefits.

At the same time, we deeply realize that if the policy selection method and the selected policy are only suitable for one region and cannot be adopted by other similar regions, it will be one-sided.



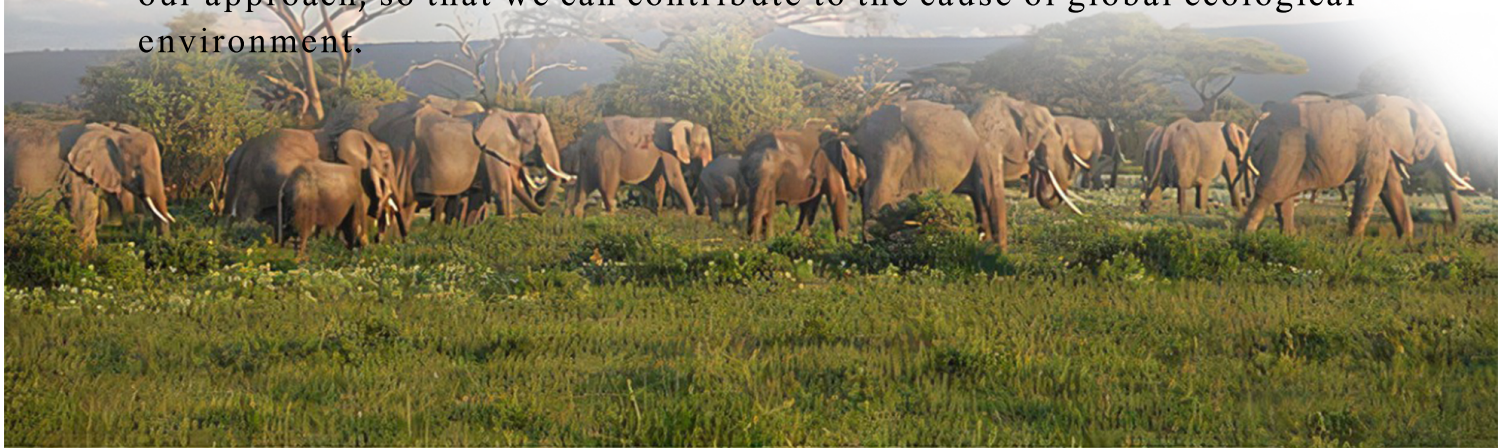
Therefore, our model takes into account this traditional disadvantage and innovatively integrates the model into random forest algorithm to automatically select the optimal policy according to the final value.

First of all, you need to search for human or wildlife related data (it is best to select accurate data in consecutive years) in the wildlife reserves where you want to make policies, and divide them according to the regions where the local pillar industries are located. For each region, you need to formulate reasonable and feasible policies according to its characteristics and corresponding indicators.

After formulating the policy, you need to collect some data and indicators related to the policy. After that, you need to control variables and change only one indicator corresponding to a policy at a time. After that, topsis and entropy weight methods are used to comprehensively evaluate the results of all policies and select the policy with the best results.

In order to be more convenient to you, we generalize this policy selection mechanism with the help of random forest model. We set up a classification model, taking all indicator data as variables input to the model, and corresponding policy choices as classification results. Finally, a classification model based on random forest is trained. The relevant model has been open source on {https://github.com/Junital/RandomForest-for-Strategy-Choosing}, which is convenient for your local wildlife protection area to learn and apply.

We have also carefully checked all the policies and management regulations formulated, and we are confident that these policies can be used by other wildlife protection bases. We hope that you can accept our approach, so that we can contribute to the cause of global ecological environment.



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# Appendices

## Appendix1: Linear Regression Model

```
import pandas as pd
from sklearn.linear_model import LinearRegression
data = pd.read_csv('data.csv', delimiter='\t')
X = data.iloc[:, :-2]
y = data.iloc[:, -2:]
print(X)
model = LinearRegression().fit(X[:4], y[:4])
X_test = X[4:10]
predicted_y = model.predict(X_test)
print(predicted_y)
```

## Appendix2: Random Forest Model

```
import numpy as np
import pandas as pd
from sklearn.datasets import load_iris
from sklearn.ensemble import RandomForestClassifier
import pickle
iris = load_iris()
df = pd.read_excel(r'Random_forest.xlsx')
fore1 = df.iloc[:, 1:10]
print(fore1)
y1 = df.iloc[:, 12]
print(y1)
clf1 = RandomForestClassifier(n_jobs=2, random_state=0)
clf1.fit(fore1, y1)
print(clf1.predict(fore1.iloc[0:6, :]))
fore2 = df.iloc[:, 1:10]
print(fore2)
y2 = df.iloc[:, 13]
print(y2)
clf2 = RandomForestClassifier(n_jobs=2, random_state=0)
clf2.fit(fore2, y2)
print(clf2.predict(fore2.iloc[0:6, :]))
print()
fore3 = df.iloc[:, 1:10]
print(fore3)
y3 = df.iloc[:, 14]
print(y3)
clf3 = RandomForestClassifier(n_jobs=2, random_state=0)
clf3.fit(fore3, y3)
```

```
print(clf3.predict(fore3.iloc[0:6, :]))
```

Appendix3: LSTM Model

```
import numpy as np
import pandas as pd
from keras import optimizers
from keras.models import Sequential, Model
from keras.layers.convolutional import Conv1D, MaxPooling1D
from keras.layers import Dense, LSTM, RepeatVector,
   TimeDistributed, Flatten, Dropout
from sklearn.metrics import mean_squared_error
from sklearn.model_selection import train_test_split
import matplotlib.pyplot as plt
import os
df = pd.read_excel("lstm_data.xlsx")
print(df)
X_train = df.iloc[:, 0]
print(X_train)
y_train = df.iloc[:, 15]
print(y_train)
X_train = np.expand_dims(X_train, axis=1)
print(X_train.shape)
model_lstm = Sequential()
model_lstm.add(LSTM(units=64, input_shape=(X_train.shape[0],
   X_train.shape[1])))
model_lstm.add(Dropout(0.4))
model_lstm.add(Dense(1))
model_lstm.compile(loss='mse', optimizer='adam',
   metrics=['mean_squared_error'])
model_lstm.summary()
history_lstm = model_lstm.fit(X_train, y_train, batch_size=10,
   epochs=50)
plt.plot(history_lstm.history['loss'], color='b', label='Training
   Loss')
plt.legend(loc='best', shadow=True)
plt.show()
test = [2030]
test = np.expand_dims(test, axis=1)
print(model_lstm.predict(test))
```